Data Quality Assessment

1. Data-Quality Assessment

1.1 Introduction

Data quality is essential to reliable business intelligence because the accuracy of dashboard insights depends on the quality, consistency, completeness, and reliability of the underlying data.

For this hospital operations analysis, the available datasets were examined and transformed to support the analysis of hospital encounters, procedures, encounter costs, payer coverage, patient demographics, and operational trends.

The assessment considers the suitability of the available data for analytical reporting and identifies areas requiring further validation before management makes operational or financial decisions.

1.2 Data-Quality Dimensions

Data-quality dimension	Assessment and implications
Completeness	The dashboard contains information on encounters, procedures, patient demographics, encounter classes, costs, payer coverage, and dates. However, the screenshots alone cannot establish whether all source records contain complete values. Missing patient identifiers, encounter dates, payer information, or cost values could affect analysis.
Accuracy	The dashboard displays measurable values, including 27,891 encounters, 47,701 procedures, and approximately $101.514M in total encounter claim costs. These figures should be reconciled against the underlying source data before being treated as independently verified totals.
Consistency	Consistent naming and categorization are important when comparing encounter classes, procedure descriptions, gender, and payer groups. Standardization should be checked to prevent duplicate categories or inconsistent classifications.
Uniqueness	Patient IDs, encounter IDs, and procedure records should be checked for duplicate entries. Duplicate encounter or procedure records could inflate counts and cost totals.
Validity	Date fields, cost values, encounter classifications, and demographic fields should be checked against acceptable formats and business rules. For example, encounter dates should be valid dates, and cost fields should contain appropriate numerical values.

Data-quality dimension	Assessment and implications
Referential integrity	Relationships between patients, encounters, procedures, and payer information should be checked to ensure that records are linked to the correct entities. Missing or unmatched identifiers could affect patient-level and encounter-level analysis.
Timeliness	The dashboard covers the period from 2011 to 2022. The completeness and consistency of records across these years should be examined to ensure that changes in reporting practices have not distorted the observed trends.

1.3 Identified Data-Quality Concerns

The following issues require particular attention:

1. **Patient and encounter identification:** Patient IDs and encounter IDs should be unique where expected. Repeated IDs must be distinguished from legitimate repeat visits to avoid removing valid records.
2. **Missing payer information:** The dashboard classifies approximately 48.71% of encounters as having no cover. This may represent encounters without recorded payer coverage, but further investigation is required to determine whether these are genuinely uninsured encounters or records with incomplete payer information.
3. **Procedure-cost interpretation:** The dashboard reports approximately $105.52M in total procedure base costs. The underlying calculation and cost definitions should be confirmed to ensure that costs are not duplicated or incorrectly aggregated.
4. **Encounter-duration units:** The dashboard displays an average visit duration of 0.18. The unit of measurement is not identified in the screenshot, so it should be confirmed before drawing conclusions about the average length of hospital visits.
5. **Historical data consistency:** The dashboard covers 2011–2022. Changes in data collection, coding practices, or reporting systems over this period could affect comparisons between years.
6. **Payer coverage and reimbursement:** Payer coverage should not automatically be interpreted as actual payment received. Claims, approved amounts, denied claims, and payment records would be required to assess actual reimbursement performance.

1.4 Data-Quality Improvement Measures

To improve the reliability of the analysis, the hospital should:

- Check for missing values in important fields such as patient ID, encounter ID, encounter date, procedure description, and cost.
- Identify and investigate duplicate records before removing them.
- Standardize categorical values such as gender, encounter class, and payer classification.
- Validate date formats and ensure that encounter and procedure dates fall within the expected reporting period.
- Check that numerical cost fields contain valid values and that calculations use the correct aggregation level.
- Verify relationships between patient, encounter, procedure, and payer records.
- Reconcile dashboard totals against source records and independently calculated control totals.

> **Assessment conclusion:** The dashboard provides a useful overview of hospital activity and expenditure. However, further source-level checks are necessary to establish the completeness, accuracy, and consistency of the underlying data. Particular attention should be given to payer classifications, patient and encounter identifiers, procedure-cost calculations, and historical data consistency.